



NATIONAL AND KAPODISTRIAN UNIVERSITY OF ATHENS

**SCHOOL OF SCIENCES
DEPARTMENT OF MATHEMATICS**

Undergraduate Statistics Program | Principal Component Analysis in R

Case Study: PCA for Retail Market Entry Strategy

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Introduction

As our retail company wishes to expand its presence in more European cities, the need for a coherent strategy is critical for taking an informed decision. To support this, we have assembled a comprehensive data set, originating from official Eurostat data. This includes key information about the demographics, economies, infrastructure and consumer basis of each city.

However, evaluating cities with 27 different metrics simultaneously may be misleading. Many of these metrics are heavily correlated, meaning some city characteristics may be overrepresented in our final evaluation. To overcome this, we will implement the Principal Components Analysis (PCA) in our dataset to reduce its dimensionality. In the chapters that follow, we will present observations, statistics, explorative graphs and statistical insight to conclude to a data-driven and strategic business decision.

Section 1: Data Exploration

Before starting the core analysis, we had to extend the data cleaning process. A simple boxplot revealed extreme outliers in the city's population domain of which we came to realize the countries of France and Romania have not been filtered out initially (on account of them not being cities). Those were removed from our dataset.

Variability and Scale Differences

A descriptive analysis revealed extreme differences in variance of our variables. For instance, the variable of "Total nights spent in tourist accommodation establishments" showed a standard deviation of 5596876.13, while "Activity rate" only 4.05. Differences of this scale, without standardizing the data, would cause large-deviating variables to completely dominate our analysis in our next steps.

Correlations and Redundant Information

Our analysis confirmed that many of the 27 indicators described similar concepts. The correlation matrix shown below (Figure 1) displays highly correlated clusters of variables, indicating a significant amount of redundant data.

To statistically validate our approach, we performed 2 robust diagnostics:

- **Bartlett Test of Sphericity:** Yielded a p-value < 0.001 , thus rejecting the null hypothesis of uncorrelated variables.
- **Kaiser-Meyer-Olkin (KMO) measure:** Achieved a score of 0.885 ("Meritorious").

Both tests confirm that our dataset is highly suitable for dimension reduction techniques.

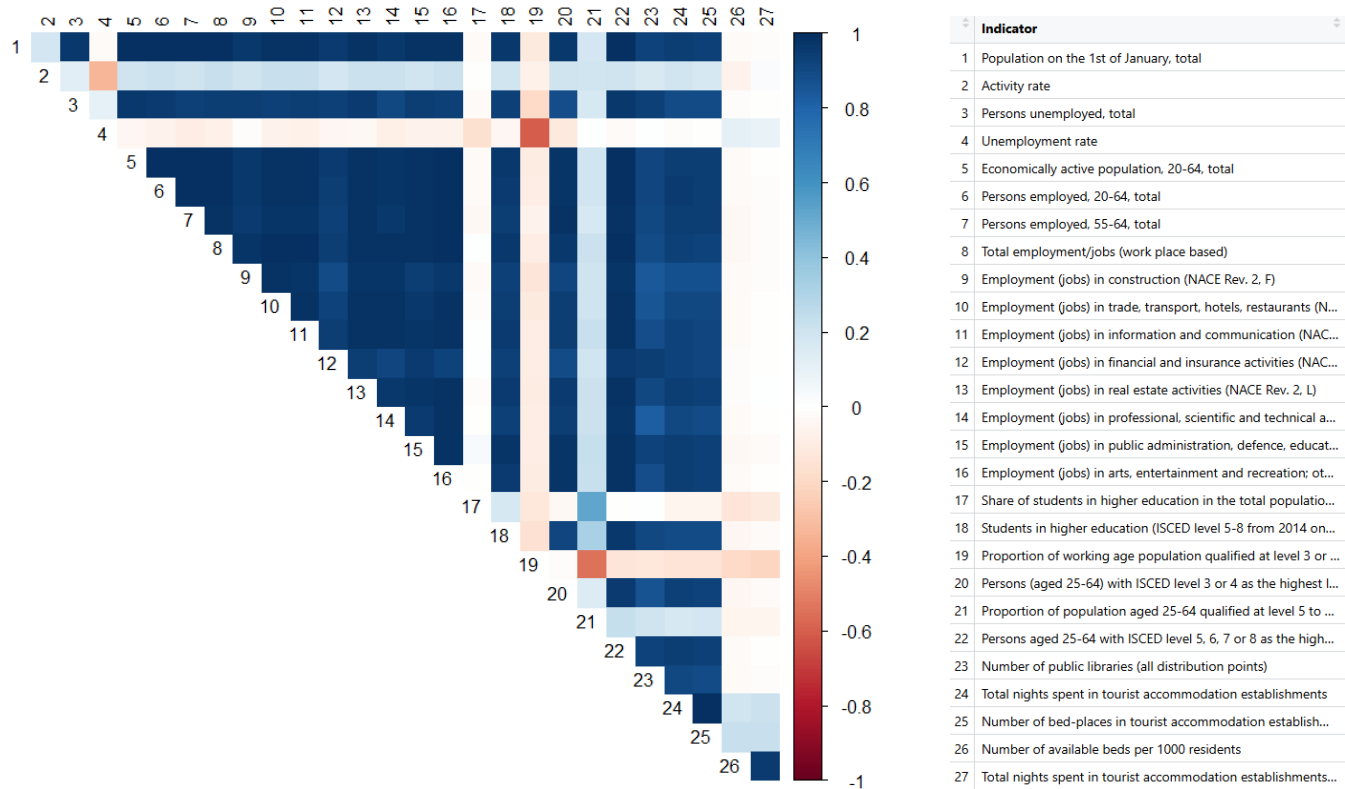


Figure 1: Correlations between the variables. Deep blue (or red) colors indicate strong positive correlation (negative respectively).

Section 2: Principal Components Selection

To identify the core factors behind market attractiveness, we performed the Principal Components Analysis (PCA) on the standardized data. This approach ensures that all the data, despite their units of measurement, contribute equally to the model.

The PCA successfully condensed the 27 variables into a smaller set of principal components that describe the vast majority of market variance. Just 3 of them include:

- The first principal component (PC1) alone describes 71.2% of total market variance.
- The second principal component (PC2) captures an additional 8.4%.
- The third principal component (PC3) for an additional 7%.
- Cumulatively, the first 3 account for 86.6% of the total market variance.

The Scree Plot

The following plot (Figure 2) depicts the proportion of market variance described by each component. By noticing the change in slope and the levelling off in the graph, it is apparent that a very small number of principal components can capture most of the relevant market data.

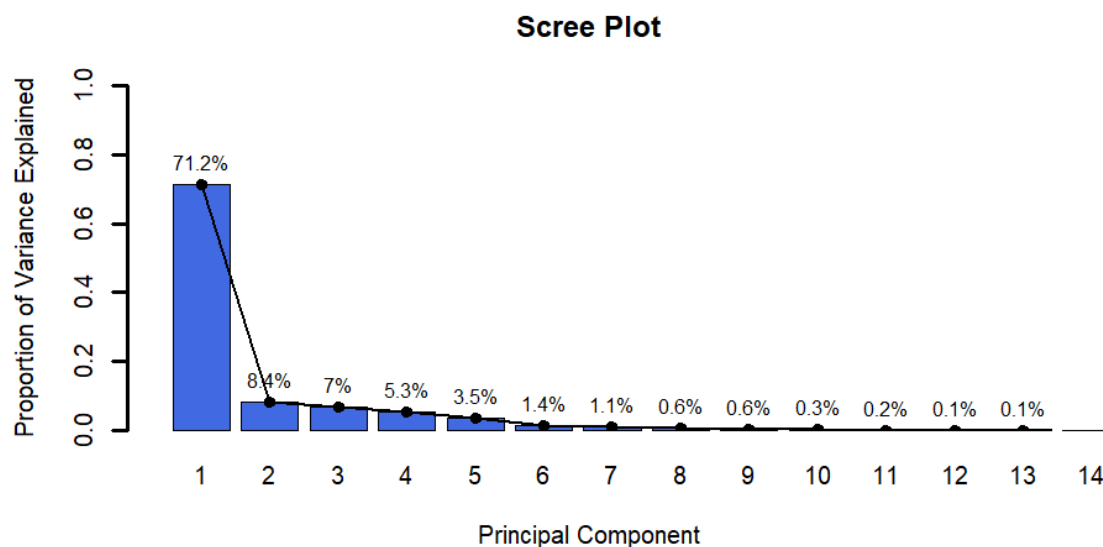


Figure 2: The scree plot. We notice that after PC5 the percentage of explained variance by the PCs becomes negligible leading us to the analysis that follows

At this stage, a critical decision must be made. Retaining all 27 principal components (PCs) would render our analysis up to this point futile. Ideally, the goal is to identify a small number of PCs that would explain substantial proportion of the total variance. As observed in the scree plot, the optimal cutoff point is ambiguous. It depicts how the first PC dominates this aspect, but the ones that follow show a small, gradual descent. For this reason, we proceeded towards 2 additional criteria:

- **The 80% rule:** We use as many PC's as needed for their sum to exceed 80% of total variance. It yielded **3 PCs**.
- **The Kaiser Criterion:** We choose all the eigenvalues greater than 1. It yielded **4 PCs**.

With those in mind, we choose to keep just the first **3 PCs**. The Kaiser criterion, as most of the time, is proven to overestimate the PCs needed and is quite strict. This choice renders our model more easily interpretable towards our client and improves the computational efficiency of our model.

Section 3: Business Interpretation of the first 2 PCs

Having concluded on using 3 PCs it is important to identify and interpret these mathematical concepts into tangible business insight. By examining the loadings (the “weight” of each variable into our component), we can identify what each component represents and to what degree on the final indicator. The table below reveals some very interesting findings about PC1 & PC2. It is worth noting that PC3 shares similarities with PC2, albeit with heavier loadings on working population metrics.

Table 1*Heaviest Loadings of the First 2 PCs*

Market Size Indicator	Loading	Tourism Indicator	Loading
Econ. active population	-0.228	No. of beds / 1000 residents	-0.593
Persons Employed	-0.228	Total nights in tourist establishments	-0.588
Popul. on the 1 st of January	-0.227	Prop. of mid-level workers	0.356
Total Employment/Jobs	-0.227	Unemployment rate	-0.327
People of the highest education level	-0.227	Total number of beds-places in tourist establishments	-0.141

Note. The first principal component is heavily influenced by absolute demographics and employment metrics. All share almost the same weight; it is meant to measure the scale of each city. The negative signs are arbitrary. The second component highlights the tourism intensity of a city. High negative scores define a major tourist destination. It also negatively correlates with unemployment, meaning that strong tourism is linked to better employment.

City Performance and Market Rankings

Based on the calculated scores from these primary 2 components we can identify some clear “winners” from each one; a group of cities with exceptional baseline customer numbers and another with dynamic tourism potential driving record profits. Those are:

- **Market Giants (PC1):** The model found Paris, Madrid and Berlin to be the absolute largest markets by scale, with absolute scores of 57.908, 26.752 & 19.312 respectively. Entering those markets guarantees access to a steady stream of customers all year round and brand recognition. Although, it should be noted that competition and operational costs are bound to be high.
- **Tourism Hubs (PC2):** Our secondary component found Benidorm & Torremolinos (Spain) and Fréjus (France) at the very top. As previously noted, those offer huge seasonal profit margins and healthy local workforce as well.

It is obvious that our future expansion strategy should not rely on a single metric. Choosing wisely from both groups guarantees all the above-mentioned benefits.

Section 4: Grouping the evaluation criteria

Based on the PCA loadings previously discussed, the picture is very clear; all 27 criteria group together on to 2 dimensions:

- **Market Scale dimension:** Variables such as Total population, Total employment, economically active population form a uniform group under PC1. They all share same loadings, unambiguously grouping together to express the demographic scale of a city.

- **Tourism economy dimension:** Criteria such as Available beds, Nights spent in tourist accommodations group under PC2, especially after considering their high loadings on this component

Ambiguous / Mixed information

While the above-mentioned variables are remarkably clear in terms of grouping, there are still some that do not fit perfectly into a single category. For example, the unemployment rate appears in PC2 (with a negative correlation with the rest) while being inherently tied to the labor force, primarily shown in PC1. Because it is influenced by both the economic scale of a city and the tourism sector, it provides mixed signals regarding its place.

Section 5: City Grouping and Regional Patterns

To explore whether cities form meaningful geographical or functional groups, we mapped their component scores (PC1 and PC2). As visualized in Figure 3 below, the cities do not distribute randomly; they form clusters that reveal underlying economic facts.

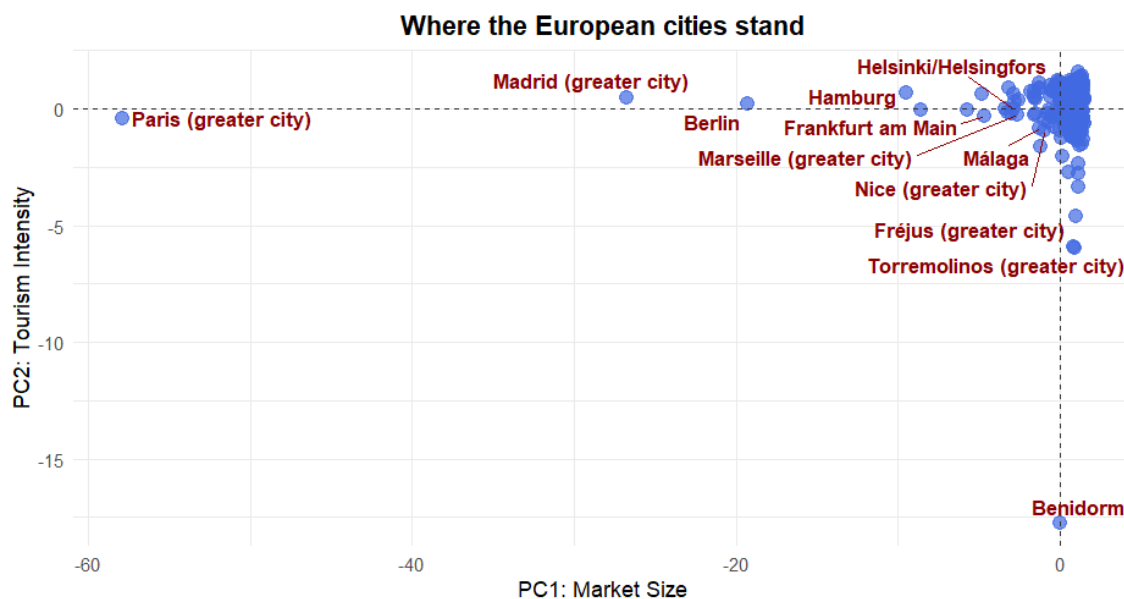


Figure 3: Scatterplot of all the cities in our dataset on the PC1 and PC2 score axis.

Some obvious functional groupings that appear are:

- **Market Giants:** On the extreme left, massive metropolis like the ones that top PC1 (Berlin, Paris, Madrid).
- **Tourism Hubs:** Benidorm, Torremolinos, Fréjus along the vertical axis.
- **Baseline cities:** Most European cities, they form a big cluster on the point of origin.

Northern vs Southern European cities

While examining regional dynamics, some separation is visible; however, we risk adding personal bias and overclaiming in our model. My selection of reference cities was not random; they intend on showing such concerns. While southern cities (Marseille, Malaga, Nice), due to favorable weather conditions, tend to be tourist hubs, a big part of them fall on the baseline category. The same can be said about the Northern European (N.E.) cities. Economic powerhouses like Frankfurt (an industrial zone), Hamburg (an international port) and Helsinki (tech sector) fit in the Market giants category, but that cannot be said for most N.E. cities, which cluster at the center. The strategic takeaway from all this is that a “one-size-fits-all” approach is bound to fail. Our model shouldn’t rely greatly on geographic constraints and look at the bigger picture; the actual economic profile of each city.

Section 6: Composite Market Attractiveness Ranking

To rank the cities from the most to least attractive, as we explored before, we cannot rely on a single metric. Therefore, we propose a defensible composite ranking method based on the 3 retained Principal Components. Each one got added in our model, multiplied by the proportion of explained variance it represents, thus creating a variance-weighted sum. This was a fair choice considering PC1 explained vastly more than the rest combined. Also, scores were inverted so that higher scores would indicate higher market attractiveness.

Table 2

Ranking Results.

Top 6 Cities	Score	Bottom 6 Cities	Score
Paris (greater city)	41.33	Saint-Louis-Bale	-0.97
Madrid (greater city)	18.95	Hayange	-0.98
Berlin	13.84	Gorlitz	-0.98
Hamburg	6.77	Frankfurt (oder)	-0.99
Munich	6.11	Greifswald	-0.99
Lyon (greater city)	3.95	Sankt Augustin	-1.02

Note. The cities on the left include the highest ranking based on our index. They dominate the market-scale component (PC1), while also having other strong economic indicators such as workforce and tourism. On the right are the 6 lowest ranking cities. They represent the markets with the smallest demographics, economies and expansion prospects in general.

Advantages and limitations of PCA for ranking

Using PCA for creating a composite index offers many advantages. It is a purely mathematically based and objective approach to take data-driven decisions. On occasions such as ours, this approach eliminates correlation bias imported from redundant variables by synthesizing uncorrelated components.

However there are inherent limitations. This whole time we focused on the most efficient way to maximize the interpreted variance without it being exactly equivalent to business value. Also, each city's calculated Composite Score is not transparent about what attribute led and to which degree to this rating.

Section 7: PCA on non-standardized data

To test our model's robustness we rerun the PCA using centered only data (center = TRUE, scale = FALSE), resulting in completely different outcomes:

- **Variance and Loadings:** The PC1 now interprets almost 100% of variance. Variables with massive absolute values dominate the loadings (Population, Total jobs etc), while others with business value such as Unemployment rate completely disappear.
- **Grouping:** The market significant Tourism Intensity dimension also disappears. City grouping seems to transform into purely demographic groups.

Conclusion: Why is standardization necessary? Our dataset mixes different types of data and measuring scales. Without the standardization, the model unjustly compares numbers well into the millions with small percentages leading to equalizing large numbers with business importance. This approach hinders our ability to recognize patterns and import them to our model.

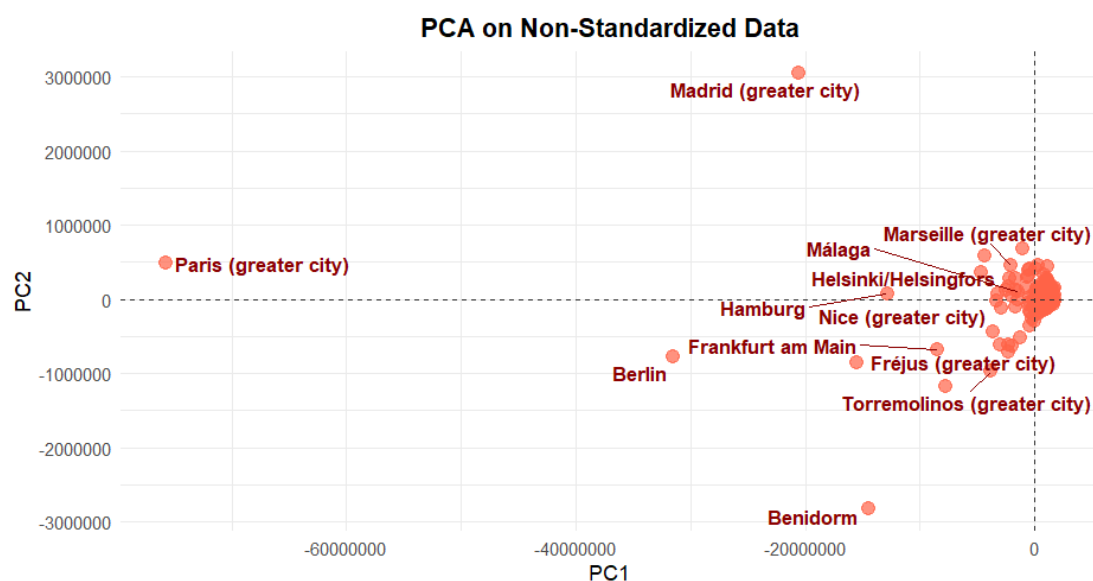


Figure 4: Notice how PC1 is dominated by large absolute values and outliers dictate the variance.

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